Bintang Dwi Marthen

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Research Interests

Systems-for-ML, ML-for-systems, operating systems, cloud computing, storage systems, and distributed systems.

Education

Bandung Institute of Technology — *Bandung, Indonesia*
MS in Computer Science — *Sep 2025 – Exp. Jul 2026*

- #1 Engineering School in Indonesia (article)
- GPA: 3.85/4.0 (study report)
- **Coursework:** Advanced Distributed Systems, Advanced Operating System, Cloud Computing Architecture, Deep Learning

Bandung Institute of Technology — *Bandung, Indonesia*
BS in Computer Science — *Aug 2021 – Jul 2025*

- #1 Engineering School in Indonesia (article)
- GPA: 3.89/4.0 (Major GPA: 3.86/4.0) (transcript)
- **Coursework:** Computer Organization and Architecture, Operating System, Computer Networks, Parallel and Distributed Systems, Distributed Application Architecture, Machine Learning, Database Management

Publications

Clock2Q+: A Simple and Efficient Replacement Algorithm for Metadata Cache in VMware vSAN — Planned for VLDB 2026

Yiyan Zhai, **Bintang Dwi Marthen**, Sarath Balivada, Vamsi Sudhakar Bojji, Eric Knauft, Jitender Rohilla, Jiaqi Zuo, Quanxing Liu, Maxime Austruy, Wenguang Wang, Juncheng Yang

Preprint PDF

Research Experience

Serverless Platform for Development Workload — *Harvard University*
International Research Collaboration — *Apr 2025 – Present*

- Collaborating with Assistant Professor Juncheng Yang (Harvard University) on designing a developer-friendly serverless platform to eliminate idle resource waste inherent in VM-based development workflows.
- Analyzed fine-grained execution traces from thousands of workloads to systematically characterize latency and resource inefficiencies in emerging serverless environments.
- Discovered that container image construction dominates startup overhead, accounting for up to 82% of total initialization time.
- Initial prototyping of optimized base image synthesis achieved up to 99.6% reduction in cold-start initialization time.

Feasibility of Static Parameterization in Adaptive Caches — *Harvard University*
International Research Collaboration — *Mar 2025 – Present*

- Collaborating with Assistant Professor Juncheng Yang (Harvard University) on re-evaluating how much adaptivity is actually needed in cache management.
- Demonstrated through extensive simulation that statically parameterized cache policies can outperform their adaptive counterparts by up to 20% in hit ratio.
- Uncovered that adaptive cache performance depends on the granularity match between the adaptation mechanism and the workload's phase behavior; mismatches in either direction lead to performance loss.
- Investigating phase-change and workload drift detection mechanisms to enable coarse-grained, stability-aware adaptivity with minimal runtime and tuning overhead.

System and AI Research (SYAIR) Training Program — *University of Chicago*
Research Training Program — *Jan 2025 – Jul 2025*

- Selected as one of the top 50 students in Indonesia for a research training program led by Prof. Haryadi S. Gunawi (University of Chicago).
- Reproduced the *Clock-PRO* cache replacement algorithm as a Chameleon Trovi artifact.
- Studied over 30 research papers from top systems conferences (*SOSP*, *OSDI*, *NSDI*, *FAST*, and others).

Sentiment Analysis Using LLMs for Twitter Data — *Bandung, Indonesia*
Undergraduate Researcher — *Oct 2024 – Dec 2024*

- Collaborating with Ayu Purwarianti (Bandung Institute of Technology) and Achmad Imam Kistijantoro (Bandung Institute of Technology) on leveraging large language models for sentiment analysis of Indonesian Twitter posts.
- Fine-tuned pre-trained transformer models and evaluated domain adaptation strategies for low-resource Indonesian text.
- Maintained and optimized NVIDIA DGX servers for large-scale model training.
- Built an internal web interface to streamline prompt testing and result comparison across multiple LLMs.
- Optimized source code, accelerating the experimentation process by approximately 10×

Work Experience

Research Intern — *Bandung, Indonesia*
Formulatrix — *Jun 2024 – Sep 2024*

- Designed and implemented a proof of concept for the automatic map discovery feature of the ROVER product, reduced configuration from three weeks to three days for non-technical users
- Introduced Simultaneous Localization and Mapping (SLAM) using a monocular front-facing camera, reducing camera count from two to one and cutting manufacturing costs by approximately 4.4%
- Modified OpenCV ArUco to support STag markers in singular and composite configurations, improving marker flexibility for prototype testing

Highlighted Projects

Simple x86 32-bit Operating System — *github repo*

- Developed a basic operating system kernel from scratch, with support for interrupts, keyboard input, and FAT32 file system management
- Implemented key file system operations such as `ls, cd, mkdir, whereis, cat, rm, cp, and mv`, providing full CRUD functionality
- Developed an electronic classroom where multiple users can simultaneously view and draw on a "chalkboard" with each person's edits synchronized

Clock-PRO Cache and CACHEUS Fix — *github repo*

- Implemented Clock-PRO cache algorithm on libCacheSim, a high-performance library for cache simulation
- Debugged and fixed faulty CACHEUS implementation

Technical Skills

Languages: C++, C, Python, Go, C#, SQL, JavaScript, Kotlin, Java

Machine Learning: vLLM, Ollama, Transformers, PyTorch, TensorFlow, Scikit-learn, Pandas

Databases and Systems: SQLite, MySQL, PostgreSQL, MongoDB, Redis, CouchDB

Systems: RabbitMQ, FEMU, libCacheSim

Cloud Computing: Google Cloud, AWS, Chameleon Cloud, CloudLab, Amazon S3

DevOps: Kubernetes, GitHub CI/CD, containerd, firecracker microVM, Docker

Misc: LaTeX, Make, CMake

References

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